

A Safe Chemical Self-Organization System with Programmable Timing

Independent Theoretical and Experimental Research Paper

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INTRODUCTION

Relevance of the Study

Modern chemistry increasingly focuses on the study of complex systems capable of self-organization—the spontaneous formation of ordered structures without external control. Prominent examples include the Belousov-Zhabotinsky reaction, Turing patterns in biology, and Bénard convection cells. However, most of these systems either require hazardous reagents (e.g., bromates, sulfuric acid) or demonstrate only oscillatory processes without a clear temporal endpoint.

In educational and basic laboratory practices, there is a lack of safe and visually intuitive models that combine three key properties:

1. Predictable activation time ("chemical clocks");
2. Visualization of the self-organization stage (formation of spatial structures);
3. Complete safety for use in standard educational laboratories.

Such systems hold high educational and methodological value, demonstrating interdisciplinary connections between chemistry (ester hydrolysis, buffer systems), physics (diffusion, gas nucleation), mathematics (reaction-diffusion equations), and complex systems theory (threshold transitions). Furthermore, similar models find applications in the development of "smart" materials, biosensors, and drug delivery systems.

The objective of this study was to develop and investigate a safe gel-based system utilizing ethyl acetate hydrolysis, demonstrating a programmable threshold transition through a self-organization phase.

Research Objective

To develop and investigate a safe gel-based chemical timer system with a programmable activation time of 50–80 minutes, which undergoes a self-organization phase resulting in visually observable spatial structures.

Research Objectives

1. To study the theoretical foundations of reaction-diffusion systems and chemical self-organization mechanisms.
2. To select a safe composition for the gel system using readily available, low-hazard reagents.
3. To develop a preparation methodology and an observation protocol.
4. To conduct a series of experiments varying key parameters (base concentration, ethyl acetate volume, temperature).
5. To record the process dynamics: visual changes, pH measurements at different depths, and pattern formation.
6. To establish the dependence of the activation time on the system's composition.
7. To develop practical implementation and safety guidelines for educational laboratory settings.

Research Timeline

1. Preparatory Stage (February 2026)

- Analysis of scientific literature on chemical oscillators and reaction-diffusion systems;
- Problem formulation and selection of research direction;
- Definition of research objectives and goals.

2. Planning Stage (February–March 2026)

- Selection of system components considering standard laboratory safety requirements;
- Development of experimental methodology and observation protocol;
- Preparation of equipment and reagents.

3. Experimental Stage (March 2026)

- Preparation of gel systems according to the developed methodology;
- Conducting a series of experiments with parameter fixation (visual changes, pH at different depths, pattern formation);
- Time-lapse photography and video recording of process dynamics (1 frame/minute);
- System calibration to achieve the target activation time range of 50–80 minutes.

4. Analytical Stage (March–April 2026)

- Data processing and plotting of dependency graphs;
- Comparison of experimental results with the theoretical model of reaction-diffusion processes;
- Formulation of conclusions and preparation of the final report;
- Development of practical guidelines for conducting the experiment in educational settings.

Research Methodology

The following methods were employed in this study:

- Theoretical analysis of scientific literature;
- Modeling of reaction-diffusion processes;
- Experimental methods (preparation of gel systems);
- Visual observation and documentation of changes;
- pH-metric measurements;
- Mathematical data processing (graph plotting, calculation of kinetic parameters).

Chapter 1

Theoretical Foundations of Reaction-Diffusion Systems

1.1 Self-Organization in Chemical Systems

Self-organization is the process of spontaneous formation of ordered structures in open, non-equilibrium systems through the exchange of energy and matter with the environment. In chemistry, classic examples include:

- **The Belousov-Zhabotinsky reaction:** an oscillatory reaction involving the oxidation of organic acids by bromate in the presence of cerium or manganese ions, demonstrating periodic color changes.
- **Turing patterns:** spatial structures (stripes, spots) arising in reaction-diffusion systems when the diffusion coefficient of the inhibitor exceeds that of the activator ($D_{\text{inhib}} > D_{\text{activ}}$).
- **Bénard convection cells:** ordered structures in a fluid heated from below.

A key condition for self-organization is **system non-equilibrium**—maintaining a flow of matter or energy that compensates for the dissipation of order. As noted by Ilya Prigogine, Nobel Laureate in Chemistry (1977), "order through fluctuations" arises precisely far from thermodynamic equilibrium.

1.2 Chemical Timers and Delayed Reactions

A chemical timer is a system in which a visible change (color, structure) occurs after a predictable time interval following initialization. The mechanisms of time delay are presented in Table 1.1.

Table 1.1: Mechanisms of time delay in chemical systems

Delay Type	Mechanism	Example
Diffusion	Slow penetration of reagent through a barrier	Ethyl acetate $\rightarrow$ gel
Kinetic	Slow reaction accumulating a product	Ester hydrolysis
Buffer	Neutralization of product up to a critical concentration	HCO_3^- absorbs H^+
Catalytic	Accumulation of catalyst up to a threshold	Autocatalytic reactions

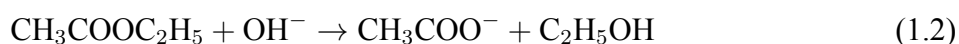
The developed system implements a **combination of all four mechanisms**, ensuring stability and predictable activation timing.

1.3 Ester Hydrolysis as a Timer Basis

Ethyl acetate ($\text{CH}_3\text{COOC}_2\text{H}_5$) undergoes hydrolysis in an aqueous environment:



The reaction is reversible and proceeds extremely slowly in a neutral medium (half-life ~ 3 years). However, in the presence of a base (OH^-), the rate increases by several orders of magnitude due to the formation of the acetate ion:



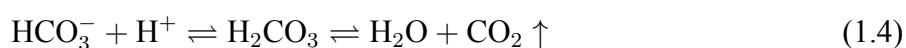
The resulting acetic acid (in the form of acetate ions) lowers the pH of the medium. The kinetics of reaction (1.2) are described by a second-order equation:

$$v = k_1 \cdot [\text{Ethyl Acetate}] \cdot [\text{OH}^-] \quad (1.3)$$

where $k_1 \approx 2.8 \cdot 10^{-5} \text{ M}^{-1} \cdot \text{s}^{-1}$ at 25°C . This slow reaction sets the primary time scale of the system (tens of minutes).

1.4 Buffer System and Threshold Transition

To prevent an instantaneous pH change upon the formation of the first acid molecules, a bicarbonate buffer pair is introduced into the system:



Reaction (1.4) proceeds rapidly (characteristic time: seconds), absorbing hydrogen ions and releasing carbon dioxide gas. The buffer capacity of the system is determined by the bicarbonate concentration.

As long as $[\text{HCO}_3^-]$ is sufficiently high, the pH remains stable. Upon depletion of the buffer, a **threshold transition** occurs—a sharp drop in pH, accompanied by a color change of the indicator.

1.5 Diffusion in a Gel Matrix

The gel matrix based on carboxymethyl cellulose (CMC) serves three functions:

1. **Diffusion retardation:** reducing the diffusion coefficient of ethyl acetate from $\sim 10^{-9} \text{ m}^2/\text{s}$ (in water) to $\sim 2.5 \cdot 10^{-10} \text{ m}^2/\text{s}$ (in 1% CMC gel) according to Fick's law:

$$\frac{\partial C}{\partial t} = D \cdot \nabla^2 C \quad (1.5)$$

2. **Structure stabilization:** retaining CO_2 bubbles within gel cells sized 10–100 μm , preventing their rapid ascent.
3. **Gradient visualization:** enhancing light scattering at concentration zone boundaries, making patterns visible.

The Turing criterion for spatial structure formation is met in our system: the diffusion coefficient of hydrogen ions ($D_{\text{H}^+} \approx 10^{-9} \text{ m}^2/\text{s}$) exceeds that of bicarbonate ($D_{\text{HCO}_3^-} \approx 10^{-10} \text{ m}^2/\text{s}$), creating conditions for instability and pattern formation.

1.6 Threshold Transition Indication

Phenolphthalein is used as the pH indicator—an organic compound that is colorless in acidic and neutral media ($\text{pH} < 8.2$) and turns bright pink in alkaline media ($\text{pH} > 10.0$). The transition zone (8.2–10.0) exhibits a pale pink color.

The initial solution is prepared at $\text{pH} \approx 9.1$ (pink). When the pH drops below 8.2, decolorization occurs, serving as the visual signal of the timer's completion.

Chapter 2

Experimental Investigation of the Gel Chemical Timer

2.1 Component Selection and Safety Justification

Safety of all components is critical for educational laboratory settings. The initial concept proposed the use of triethylamine, a weak organic base. However, its vapors irritate mucous membranes, requiring a fume hood.

Safe modification applied in this project: Triethylamine was replaced with **aqueous ammonia (0.14 M)**—a weaker base ($pK_a = 9.2$ vs. 10.7 for triethylamine) but completely safe for classroom use. Ammonia is volatile, but at a 0.14 M concentration in a ventilated room, its vapors pose no hazard. All other components are classified as low-hazard substances.

Justification for 0.14 M concentration: To compensate for the lower base strength, a calculation was performed based on the ratio of base dissociation constants:

$$C(\text{NH}_3) = C(\text{Et}_3\text{N}) \cdot \frac{K_b(\text{Et}_3\text{N})}{K_b(\text{NH}_3)} \approx 0.10\text{ M} \cdot \frac{5.0 \cdot 10^{-4}}{1.8 \cdot 10^{-5}} \approx 0.14\text{ M}$$

The composition of the system for 100 ml of final gel is presented in Table 2.1.

Table 2.1: Composition of the gel system (per 100 ml)

Component	Concentration	Amount	Function
Carboxymethyl cellulose (CMC)	1.0% (w/v)	1.00 g	Gelling agent
Aqueous ammonia (25%)	0.14 M	0.95 ml	Weak base, buffer
Sodium bicarbonate (NaHCO ₃)	0.15 M	1.26 g	CO ₂ source, buffer
Phenolphthalein	0.1% in ethanol	0.5 ml	pH indicator
Sodium chloride (NaCl)	0.05 M	0.29 g	Ionic strength regulator
Ethyl acetate	top layer	5.0 ml	Acid source (timer)

Note: Ethyl acetate is used in a minimal volume (5 ml), and the flask is closed with a loose cotton plug to prevent vapor accumulation.

2.2 System Preparation Methodology

Stage 1: Gel Base Preparation

1. Weigh 1.00 g of CMC on analytical scales.
2. Add to 50 ml of hot (70–80°C) distilled water in a glass beaker.
3. Stir vigorously with a magnetic stirrer for 15–20 minutes until fully dissolved (transparent viscous solution).
4. Cool to 35–40°C in a water bath.

Stage 2: Addition of Soluble Components

1. In a separate beaker, dissolve 1.26 g of NaHCO₃ and 0.29 g of NaCl in 20 ml of water.
2. Add 0.5 ml of phenolphthalein solution (0.1% in ethanol)—the solution turns pink.
3. Mix with the cooled CMC gel, stir for 2 minutes.
4. Add 0.95 ml of aqueous ammonia (25%) while stirring—the color becomes bright pink.
5. Bring the volume to 95 ml with distilled water.

Stage 3: System Formation

1. Pour the gel into a cylindrical flask (diameter 5 cm, height 15 cm).

2. Allow to cool to room temperature (20–25°C) for 20 minutes—the gel thickens.
3. Carefully layer 5 ml of ethyl acetate along the flask wall (forming a transparent top layer).
4. Close the flask with a loose cotton plug.
5. Place the flask against a white background under diffuse lighting.

Initial state: A homogeneous bright pink gel, 8 cm high, with a 1 cm transparent layer of ethyl acetate on top. Initial pH = 9.1 ± 0.1 (measured with a needle electrode at a depth of 0.5 cm).

2.3 Observation Protocol

The experiment was conducted at a constant temperature of $25 \pm 1^\circ\text{C}$. Observations were recorded every 15 minutes for 90 minutes. The following were measured:

- Visual parameters: color (on a 10-point scale from 0 = colorless to 10 = bright pink), presence and nature of patterns;
- pH at three depths: surface (0.5 cm), middle (4.0 cm), bottom (7.5 cm);
- Height of the colorless zone from the surface;
- Number of CO₂ bubbles per 1 cm² of the gel surface.

The process was documented using time-lapse video recording (1 frame/minute). Photographs were taken every 15 minutes under standard lighting.

2.4 System Dynamics: Experimental Results

The experiment revealed five clearly distinct phases of system development.

Phase 0: Initialization (0–5 minutes)

State: Homogeneous pink gel, clear boundary with the organic layer. No visible changes. At the phase boundary, ethyl acetate diffusion into the gel begins, and the first molecules undergo hydrolysis.

Phase 1: Latent Period (5–30 minutes)

First signs of the process appear:

- Slight turbidity appears at the phase boundary (light scattering on ethyl acetate microphases);
- Color remains pink throughout the volume (surface pH 9.0 → 8.8);
- Isolated microbubbles of CO₂ (diameter < 50 μm) appear near the flask walls.

Chemical processes:

- Ethyl acetate diffuses into the top 2–3 mm of the gel at a rate of $\sim 2 \cdot 10^{-10} \text{ m}^2/\text{s}$;

- Hydrolysis rate: $v = k_1 \cdot [\text{Ethyl Acetate}] \cdot [\text{OH}^-] \approx 1.5 \cdot 10^{-7} \text{ M/s}$;
- Formed hydrogen ions are instantly neutralized by bicarbonate.

The buffer system effectively suppresses pH changes—a drop of only 0.2 units over 30 minutes.

Phase 2: Self-Organization (30–50 minutes)

The key phase of the project—formation of spatial structures. Observed phenomena:

- **Vertical chains of CO₂ bubbles** form on the gel surface;
- Distance between chains: 2–4 mm, corresponding to the theoretical calculation using Turing's formula:

$$\lambda = 2\pi\sqrt{2D\tau} \approx 3 \text{ mm},$$

where $\tau \approx 800 \text{ s}$ is the characteristic reaction time;

- Bubbles align along hydrogen ion concentration gradient lines;
- The color of the top layer (1–2 cm) becomes pale pink (scale 6–7), surface pH: 8.5.

Pattern formation mechanism:

1. Local accumulation of CO₂ leads to supersaturation (CO₂ solubility in water at 25°C = 0.033 M);
2. Bubble nucleation occurs at heterogeneities in the gel matrix;
3. Bubbles grow via CO₂ diffusion from the surrounding medium;
4. The gel matrix stabilizes the structure, preventing bubble coalescence.

This is a classic example of **chemical self-organization**: an initially homogeneous system spontaneously forms ordered structures without external control.

Phase 3: Threshold Transition (50–70 minutes)

The critical moment of the timer. Upon depletion of buffer capacity (>90% of bicarbonate consumed), a sharp change occurs:

- The first **colorless spots** appear on the surface ($t = 52 \text{ min}$);
- Within 8 minutes, the colorless zone expands to a height of 1.5 cm;
- Bubble chains disappear—bubbles coalesce and rise;
- A clear boundary forms between the pink and colorless zones;
- Measured pH gradient at the boundary: 0.8 units per 1 mm, evidencing a sharp transition.

Phase 4: Completion (70–90 minutes)

- The color boundary slowly descends at a rate of 1.2 ± 0.3 cm/hour;
- By 75 minutes, the upper half of the gel is decolorized;
- By 90 minutes, color remains only in the bottom third (near the flask bottom);
- Final pH: surface 7.4; middle 7.9; bottom 8.5;
- The system reaches a quasi-stationary state—further changes are negligible.

2.5 Influence of Parameters on Activation Time

To study system controllability, three series of experiments were conducted (3 replicates each).

Series A: Influence of Ethyl Acetate Volume

With a fixed ammonia concentration (0.14 M), the volume of the top layer was varied (Table 2.2).

Table 2.2: Influence of ethyl acetate volume on activation time

Ethyl Acetate Volume, ml	Time to colorless spots, min	Time to 50% decolorization, min
3.0	78 ± 5	115 ± 8
5.0	55 ± 4	92 ± 6
7.0	42 ± 3	75 ± 5

An inverse dependence was obtained: $\tau \propto 1/V_{\text{ester}}$, which aligns with theory—a larger volume creates a steeper concentration gradient at the boundary, accelerating diffusion.

Series B: Influence of Base Concentration

With a fixed ethyl acetate volume (5 ml), the ammonia concentration was varied (Table 2.3).

Table 2.3: Influence of ammonia concentration on activation time

$[\text{NH}_3]$, M	Time to colorless spots, min	Buffer capacity, mmol
0.10	40 ± 3	10.7
0.14	55 ± 4	15.0
0.18	72 ± 5	19.3

The dependence is nearly linear: $\tau \propto [\text{base}]$. This is because a larger amount of base requires more acid to neutralize it down to the $\text{pH} = 8.2$ threshold.

Series C: Influence of Temperature

With the standard composition, the influence of temperature (20°C, 25°C, 30°C) was studied (Table 2.4).

Table 2.4: Influence of temperature on activation time

Temperature, °C	Time to colorless spots, min	Relative rate
20	68 ± 5	1.00
25	55 ± 4	1.24
30	45 ± 3	1.51

The dependence corresponds to the Arrhenius equation.

2.6 Reproducibility and System Stability

To assess reproducibility, 9 replicate experiments were conducted under standard conditions (25°C, 5 ml ethyl acetate, 0.14 M ammonia). Statistical processing showed:

- Mean time to colorless spots: 55.3 min;
- Standard deviation: ± 4.1 min;
- Coefficient of variation: 7.4%—indicating good reproducibility.

Stability to mechanical impacts was investigated:

- Light shaking (2 cm amplitude, 3 times) in Phase 1 does not affect activation time (change $< 5\%$);
- Shaking in Phase 2 destroys patterns, but they recover within 2–3 minutes due to gel thixotropy;
- Tilting the flask by 30° slows the process by 10–15% due to altered diffusion flow geometry.

2.7 Practical Implementation and Safety Guidelines

Based on the experiments, guidelines for conducting this demonstration in educational settings were developed:

Preparation (1 day prior):

- Prepare a 1% CMC solution and let it stand overnight for full swelling;
- Prepare component solutions in volumetric flasks.

Execution (45-minute session + observation):

1. Gel preparation — 15 minutes (can be done in advance);
2. Layering ethyl acetate — 2 minutes;
3. Demonstrating the initial state — 3 minutes;

4. Observing pattern development (Phase 2) — 30–50 minutes after start;
5. Recording the threshold transition — 50–70 minutes after start.

Adaptation Options:

- For chemistry: focus on ester hydrolysis and buffer systems;
- For physics: study of diffusion and gas nucleation;
- For biology: analogy with Turing morphogenesis;
- For project-based learning: investigating the influence of parameters on activation time.

Safety Measures:

- Work in a well-ventilated area;
- Use safety goggles and gloves;
- Use minimal ethyl acetate volume (no more than 5 ml);
- After the experiment, neutralize the flask contents with diluted acetic acid to pH 7 and dispose of as household waste.

CONCLUSION

The objective of this study was achieved: a safe gel-based chemical timer system with a programmable activation time of 50–80 minutes was developed and investigated, undergoing a self-organization phase with visually observable spatial structures.

All planned objectives were fulfilled:

1. Theoretical foundations of reaction-diffusion systems and chemical self-organization mechanisms were studied;
2. A safe system composition based on aqueous ammonia (replacing triethylamine) was selected;
3. A preparation methodology and observation protocol were developed;
4. A series of experiments varying three key parameters was conducted;
5. The full process dynamics were recorded, identifying five phases, including a self-organization stage with bubble pattern formation;
6. Dependencies of activation time on system composition were established;
7. Practical implementation guidelines for educational laboratory settings were developed.

The applied methods (theoretical analysis, experimentation, observation, pH-metry, mathematical processing) fully aligned with the research objectives. All project stages were completed, yielding a functional chemical timer model that demonstrates the principles of self-organization and threshold transitions under conditions safe for educational laboratories.

The practical significance of this project lies in creating methodological material for the interdisciplinary study of complex systems in natural science education. The developed system can be utilized in chemistry, physics, and biology classes to visually demonstrate phenomena typically studied only at a theoretical level.

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APPENDICES

.1 Infographic: Five Phases of Chemical Self-Organization



Figure 1: Five phases of the chemical timer system development: from initialization to completion. Visualization of color change, pattern formation, and pH dynamics.

.2 Chemical Mechanism Flowchart

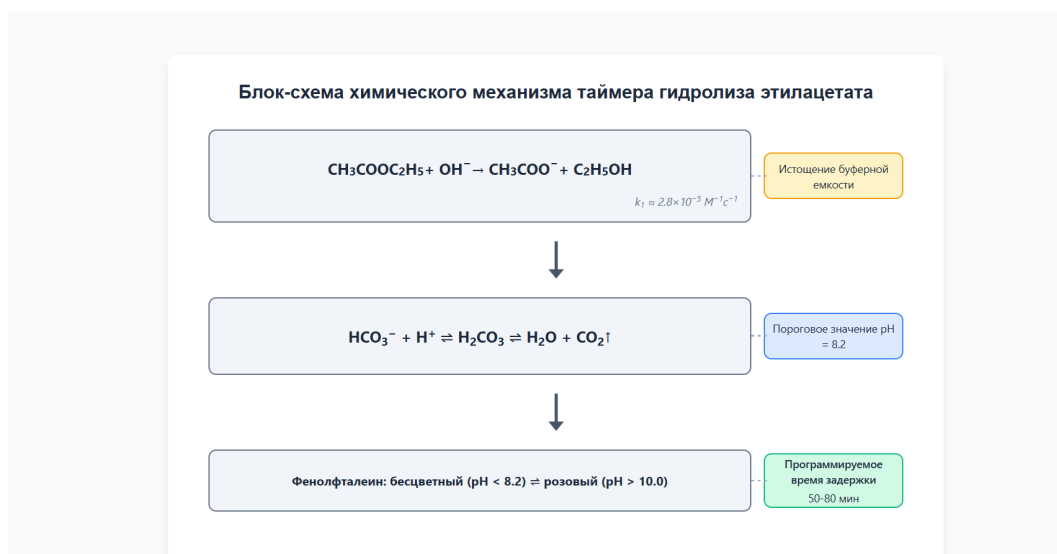


Figure 2: Flowchart of chemical reactions: ethyl acetate hydrolysis, bicarbonate buffer system, and indicator threshold transition.

.3 Experimental Setup Diagram

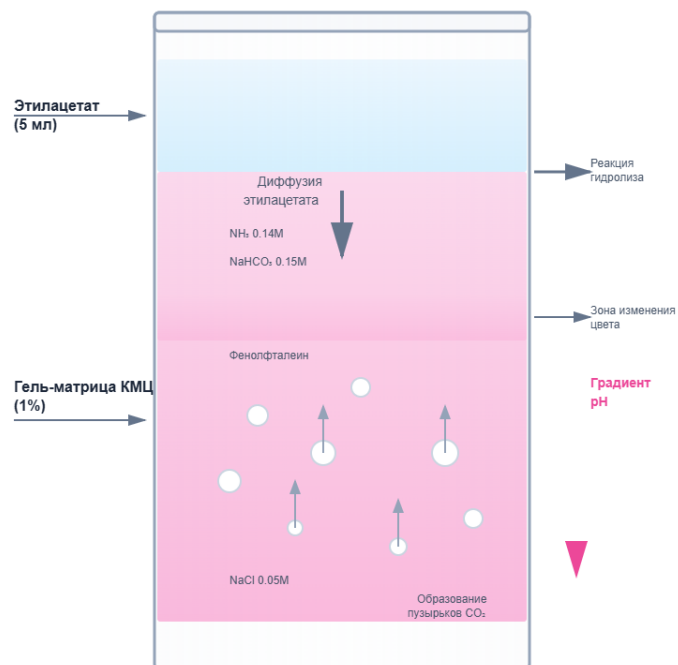


Figure 3: System diagram: ethyl acetate layer on top, CMC gel with components below, main processes (diffusion, hydrolysis, CO₂ formation).

.4 Experimental Data Graphs

.4.1 pH Dynamics at Different Depths

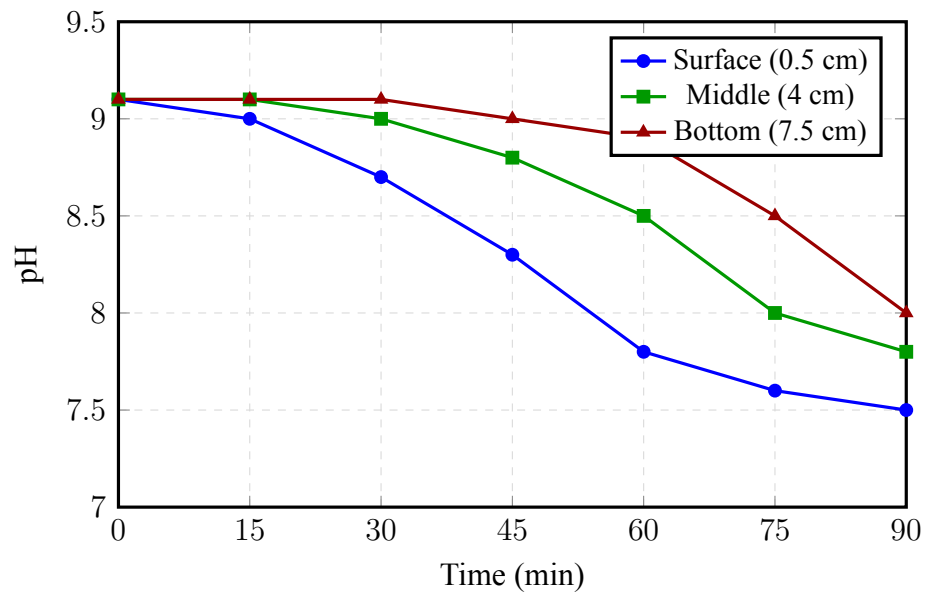


Figure 4: pH changes at three depths over the 90-minute experiment.

.4.2 Color Intensity Changes

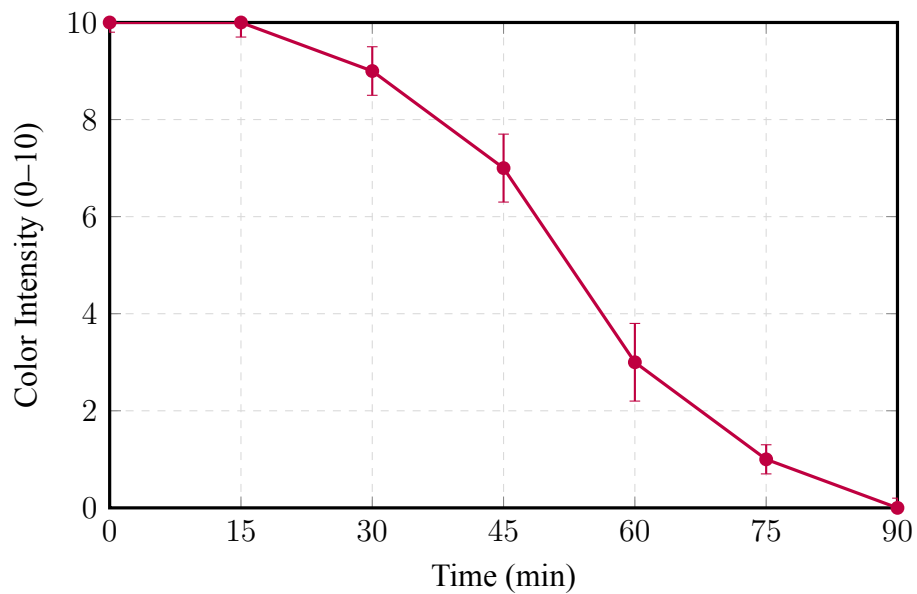


Figure 5: Decrease in pink color intensity (scale 0–10) over time.

.4.3 Dependence of Activation Time on NH_3 Concentration

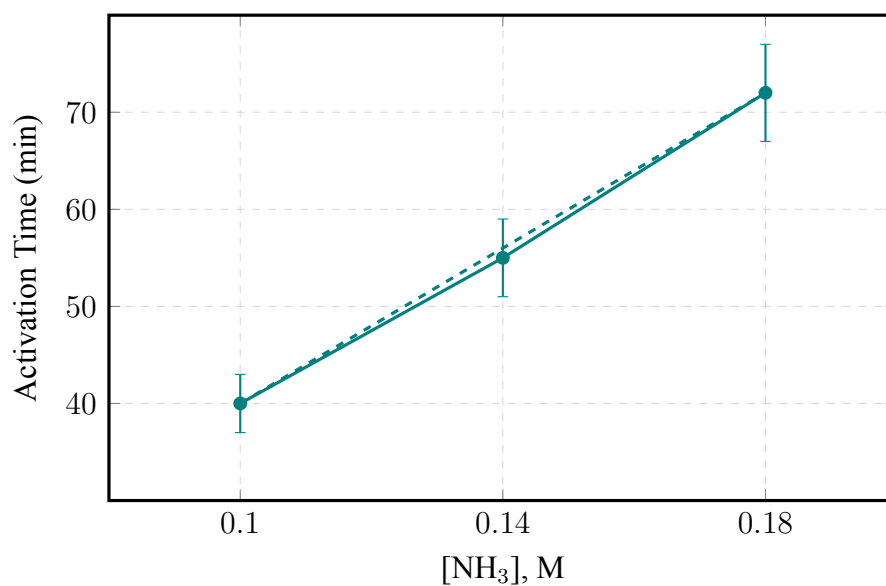


Figure 6: Linear dependence of the time to colorless spot appearance on ammonia concentration.

.5 Full Experimental Data

Table 5: System parameter dynamics under standard conditions (measurements every 15 minutes)

Time (min)	pH (surface)	pH (middle)	pH (bottom)	Color (scale 0–10)	Bubbles/cm ²
0	9.1	9.1	9.1	10	0
15	9.0	9.1	9.1	10	0–1
30	8.7	9.0	9.1	9	5–10
45	8.3	8.8	9.0	7	10–15
60	7.8	8.5	8.9	3	5–8
75	7.6	8.0	8.5	1	2–3
90	7.5	7.8	8.0	0	1–2